



◆ **LEAD AI ENGINEER**
Founder · Archi Cam AI

CONTACT & PROFILES

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AI & LLM STACK

- Google Antigravity IDE ■■■■■■
- LangGraph & CrewAI ■■■■■■
- Google Gemma 4 (12B) ■■■■■■
- Gemini 2.5 / 1.5 Pro ■■■■■■
- Google TabFM (Tabular) ■■■■■■
- FastMCP & Genkit ■■■■■■
- Neo4j GraphRAG — Agent K1

DATA & GRAPHS

- Neo4j / Cypher Graph ■■■■■■
- PostgreSQL / pgvector ■■■■■■
- Apache AGE & DuckDB ■■■■■■
- BigQuery DataFrames ■■■■■■
- Pandas / NumPy ETL ■■■■■■

DEV & MLOPS

- Python 3.11+ / MLOps ■■■■■■
- FastAPI / Next.js 14 ■■■■■■
- MLflow & Data Drift ■■■■■■
- SHAP Sentinel Audit ■■■■■■
- IfcOpenShell & Shapely ■■■■■■
- Docker & Pytest ■■■■■■

ETHICS, SECURITY & AUDIT

- ◆ CCAA Excellence Attestation '23
- ◆ AVSEC Officer (ICAO Annex 17)
- ◆ AI Ethics & Anti-Hallucination
- ◆ OKF v0.2 SHA-256 No-LLM
- ◆ EU AI Act Compliance (RSASSA)

LANGUAGES

- French — Native / Fluent
- English — Professional / Tech.

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | Founder @ Archi Cam AI

◆ EXECUTIVE SUMMARY

Lead AI Engineer & Data Architect (Google Developer Program Member), leveraging Google Antigravity to design neuro-symbolic multi-agent architectures, Neo4j GraphRAG, and sovereign MLOps pipelines. Expert in AI ethics: SHAP transparency, EU AI Act compliance, and deterministic zero-hallucination models. Founder & Architect of Archi Cam AI (Agentic AI SaaS & 5D BIM), bridging civil engineering methodology, mathematical rigor, and product vision.

◆ FLAGSHIP AI PROJECTS

Archi Cam AI — *Agentic AI & 5D BIM Construction SaaS*

- ▶ Full automation of construction quantity takeoff and cost estimation: eliminating human error and material waste.
- ▶ Instant generation of standardized BOQ estimates in <45s (99.2% time savings) and 3D architectural renders.
- ▶ Deterministic zero-hallucination structural concrete validation enforcing strict compliance with urban planning codes.

K1-MATHINFO (v3.0.0) — *Sovereign Academic Research & Asset Platform*

- ▶ Preservation, indexing, and semantic exploration across 28 years of university scientific research (470 Ph.D. theses).
- ▶ Intelligent student-to-advisor recommendation (Advisor Matcher) and cryptographic certification of research integrity.

Sovereign.BI Agentic — *Enterprise Business Intelligence & Strategic Decision Engine*

- ▶ Executive decision enablement: querying complex corporate data warehouses directly in natural language in under 5 seconds.
- ▶ Total sovereign enterprise privacy with air-gapped local execution and mathematical explainability for every KPI.

Dataset Automator & VigieSahel — *Climate Resilience & Production MLOps Systems*

- ▶ Dataset Automator: Autonomous data engineering factory for reliable dataset preparation, fairness audits, and compliance.
- ▶ VigieSahel: Sahelian resilience system cutting crop sowing failures by 35% and forecasting meningitis epidemics 14 days ahead.

◆ PROFESSIONAL EXPERIENCE

AI Lead & Data Science Consultant — *March 2026 – Present*

Independent Projects & Enterprises | Douala, CM

- ▶ Ethical sovereign AI systems: GraphRAG Neo4j pipelines, high-dimensional EDA, multi-source RAG architectures.
- ▶ Executive decision reporting, automated KPI dashboards, and SHAP explainability audits for African SMEs.

Aviation Security Officer (AVSEC) — *2018 – Present*

CCAA — Cameroon Civil Aviation Authority | Douala, CM

- ▶ Threat assessment, secure access control, and regulatory compliance audits (ICAO Annex 17).
- ▶ Operational crisis management: emergency team coordination, anti-intrusion protocols.

◆ EDUCATION & CERTIFICATIONS

M.Sc. in Applied Artificial Intelligence — *Dec. 2025 – 2027 [In Progress]*

University of Ngaoundéré | Cameroon

- ▶ ML & Bayesian Statistics, Data Engineering & Neo4j, Computer Vision & Robotics, Ethics & Cybersecurity, Production MLOps.
- ▶ Research project: K1-MATHINFO v3 sovereign system — multi-source agent certified OKF v0.2 (SHA-256 No-LLM).

B.Sc. in Civil Engineering (Building Option) — *2015 – 2016*

ISTDI / IUC Douala | Cameroon

- ▶ Structural calculations (BAEL 91), quantity surveying, construction project management — AI-applied estimation base.

◆ HONORS & APPLIED AI DISTINCTIONS

- ◆ CCAA Attestation of Excellence & Integrity (2023) — Awarded by the Director General for outstanding operational performance & ethics.
- ◆ Google Cloud #AllThingsAgentic Hackathon — Dataset Automator v4.0 (Google Antigravity, TabFM, BigQuery DataFrames, WIT).
- ◆ Google Developer Program · AICC Accra — Active member · Accra AI Community Centre & Google for Startups Accelerator Network.